

Introduction

Unit 1

Negar Mehr

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ME 132: Welcome



Negar Mehr



Seoyeon Choi



Ryan Park

Course Logistics

- 1 Prereqs: Math 53, Math 54, Physics 7A, Physics 7B, and E7
- 2 Texts: no official text, notes provided on bCourses
- 3 Reference: Astrom and Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, Princeton University Press .. advanced text, but excellent .. uploaded to bCourses
- 4 Discussions: 4 sections, [1109 Etcheverry](#)
Attend the one you are registered for
Start Tuesday, September 1st
- 5 No Physical Lab
- 6 Use of ChatGPT or other LLMs is permitted. However, you MUST declare that you have used such resources.

Course Logistics ...

- 1 ED for peer help and reaching out to GSIs... make sure you have access ... use this resource!
- 2 Office hours:
 - Negar: Mondays 11:30 am - 1pm, 5126 Etcheverry Hall
 - Seoyeon: Wednesdays 11 am - 12 pm, 1171 Etcheverry
 - Ryan: Thursdays 1 - 2 pm, 1171 Etcheverry
- 3 Lecture notes will be posted on bCourses
- 4 Visit bCourses often for announcements etc
- 5 DSP student exception letters must be received by Sept 25, 2026
- 6 **Zero cheating tolerance policy**

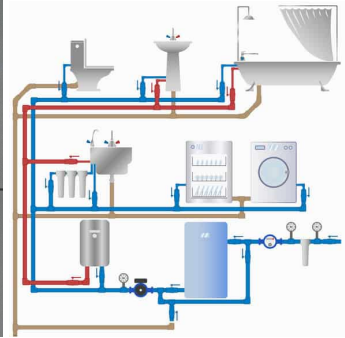
Grading

- Midterm 1: 20% weight Friday, October 9th, in class
- Midterm 2: 20% weight Friday, November 13th, in class
- Final: 50% weight Monday, December 14th, 8-11am
- Homework: 10% weight

Homework

- Issued every Friday, due Sunday of the following week by 11:59pm sharp on GradeScope
- DSP students: due following Wednesday by 11:59pm sharp
- Submit early to avoid problems uploading
- No late homework accepted ... but you get to drop 2 lowest scores
- HW0 is posted!!
- Due Sunday September 6, Wednesday September 9 (DSP students)

What is Feedback Control? Shower Example

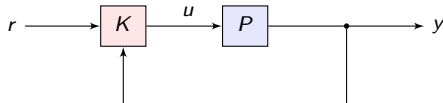
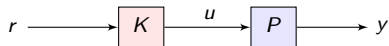


Can control shower temperature within $\pm 0.2^\circ \text{C}$!
Don't need to know details about shower system!
Works even in the presence of disturbances like flushing toilets!

Feedback Control

- What can feedback control do?
 - make a precision instrument out of inexpensive crude components
 - works reliably even in the face of disturbances and noises
- Fundamental strategy of feedback control:
 - decide what action to take based on measured information
- Limitations: Feedback control can't do everything
 - cannot magically transform a Kia into a Ferrari
 - dynamics of plant limit what control can do
 - ex: delays due to long pipes in shower example

Open-loop vs Closed-loop or Feedback Control



- Plant P , controller K
measurements y , control u
- Open loop control
 - measurements are not used to adjust control
 - strategy: calibration
- Closed-loop or feedback control
 - control adjusted continuously based on measurements
 - much more powerful
 - don't need to know P accurately
 - works in uncertain environments
 - strategy: sense, compute, actuate

Applications Abound

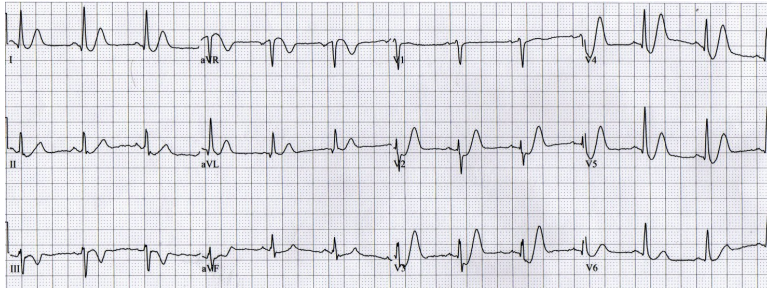
- Mechanical systems: elevators, robot end-effectors, manufacturing equipment, self-driving cars
- Aerospace systems: aerospace vehicles, drones
- Chemical systems: distillation columns, aluminum smelting, plastics, food processing
- Electrical systems: control of voltage and AC frequency across 10^5 Km
- Human behavioral systems: persuade people to buy stuff with personalized advertising feedback
- Biological systems: white blood cells track down invading viruses and bacteria
- Many many others ... economics, medicine, biology, engineering, political science ...

Watch the SpaceX video

For Whom this Course Tolls

- What you will learn
 - mathematical framework to analyze and design control systems
 - applications of control drawn mainly from mechanical systems: cars, robots, aerospace
 - understand how dynamics limit the performance of control systems
- What I hope you will do with this
 - I am only exposing you to the basic stuff
 - many much more complicated problems out there
 - self-driving cars
 - therapeutics for diabetes, Alzheimer's
 - coordinating drones for fire fighting
- I hope to inspire you to transform the lives of others using feedback control

Signals



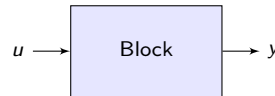
Electrocardiogram signals from a middle aged patient presenting with chest pain and diaphoresis.
Courtesy Mat Goebel.

Signals

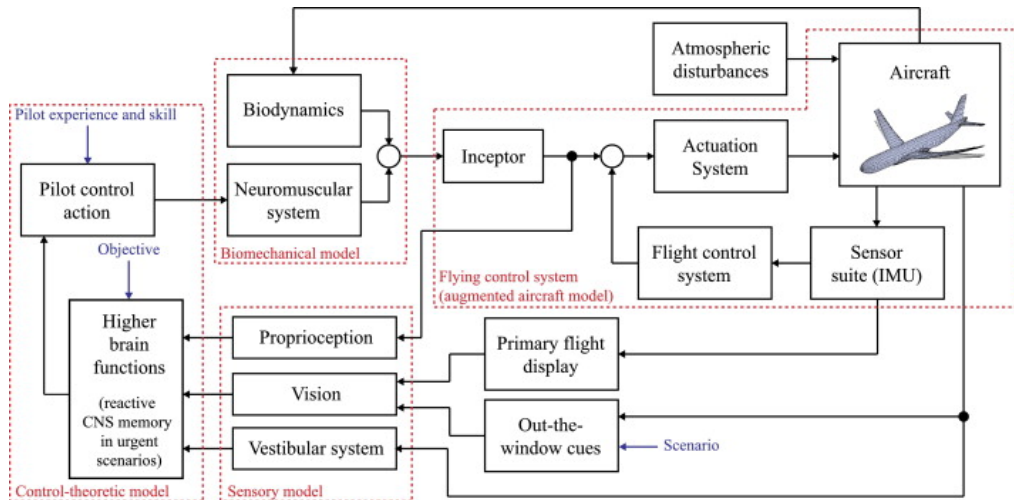
- Signals are just functions of time
 - notation: whole signal u , value of signal at time t is $u(t)$
 - represented by wires in a block diagram
- Often signals are converted to voltages
 - integrated sensors for temp, pressure, humidity, kinematic variables
 - why? because voltage signals can be manipulated easily using electronics/software
- Vector-valued signals
 - many signals can be bundled together and are represented as a single wire
ex: temperatures at 10 different locations
these are called vector-valued signals

Blocks

- Blocks take signals in and produce signals out
- Ex: Gain block, integrator block, saturation block
- Block can be combined to produce more complicated actions
- Abstract blocks: represent physical reality
- Blocks can have many inputs and/or many outputs
- Ex: airplane, combustion systems, chemical process



The Idea of a Block Diagram



Models versus Physical Systems

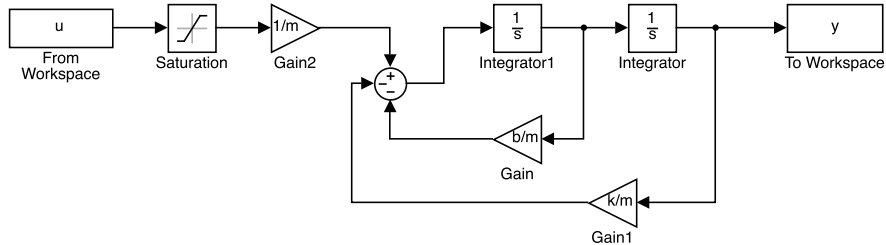
- Physical systems like airplanes are **modeled**
 - diff eqns, look-up tables, if then else logic, probability distributions
- Models
 - approximations of physical reality
 - **no model is perfect**: always have errors
- Modeling Errors
 - capture difference between model and physical reality
 - ex: could say model is $\pm 5\%$, accurate ... more complicated ways later
- Range of models
 - very accurate, complex models: used in simulation and verification
 - very simple approximate models: used in control system design

SimuLink

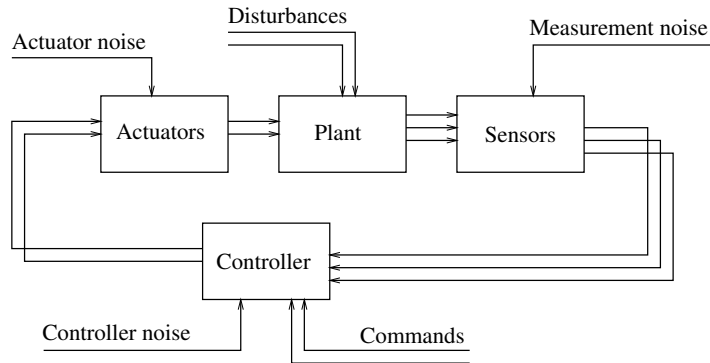
a graphical user interface tool to build and simulate models

very commonly used in industry

Boeing aircraft models will run to 5000 pages of simulink block diagrams!



Structure of Control Systems



P plant
 K controller
 u input
 y output
 d disturbance
 n noise
 r reference

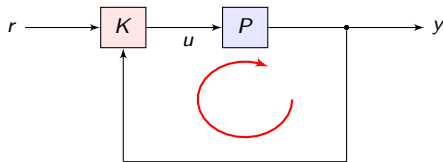
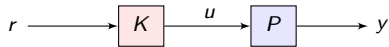
Lexicon

Plant	physical system to be controlled	ex: aircraft, robot, car, chemical plant
Sensor	device that measures some aspect of plant	ex: speedometer, GPS sensor, thermometer
Actuator	electromechanical device that can change plant behavior	ex: valve, motor
Controlled variables	variables that we wish to affect or keep small	ex: position error, velocity error, fuel consumption

Lexicon ...

Reference Command or	external signal that we want controlled variable to follow	ex: speed of car in cruise control, altitude in aircraft control
Setpoint	reference that is a constant value	ex: 65 mph, 35000 ft
Disturbances	exogenous signals that affect the plant	ex: wind gusts in aircraft lateral control, incline of roadway in car cruise control
Sensor noise	random signals that represent error in sensor readings	ex: quantization error
Actuator noise	random signals that represent error in actuator response	ex: quantization error, nonlinearities
Controller	algorithm that takes sensor signals and reference signals and produces actuator inputs	ex: could be hardware or software

Open-loop vs Closed-loop or Feedback Control

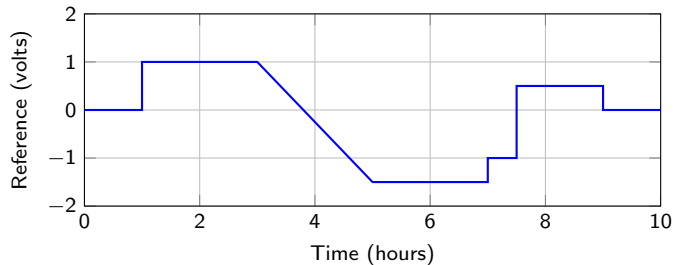


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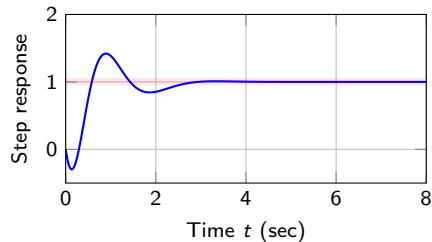
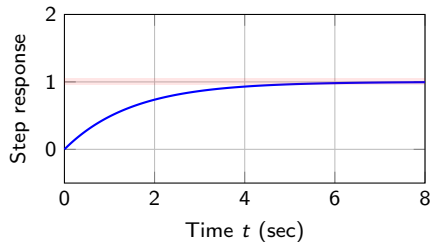
Example: reference for the elevation of an airplane

Constant reference is called a **set-point**



Performance

- Many different notions: tracking error, settling time, minimum fuel, etc
- Depends on application context
- Consider two responses to a step reference shown below:
 - which is better for an elevator control application?
 - which is better for a robotics manufacturing application?



Disturbance Rejection

- Disturbance are random signals that affect the plant behavior
- Examples
 - grade of roadway in cruise control
 - wind gusts in aerospace control
 - ambient temperature in climate control
- Disturbance Rejection: controller should deliver performance even in the presence of disturbances

Robustness

- True plant P
- Plant model P^o
- Controller K is designed using the model P^o
- Controller complexity $\approx$ model complexity
so we use a simple model for controller design
- Verification
 - of course K will work for the model P^o !
 - but it should work for the true plant P
 - so we use a detailed accurate model for verification
- Robustness: controller works even when we have significant plant model errors

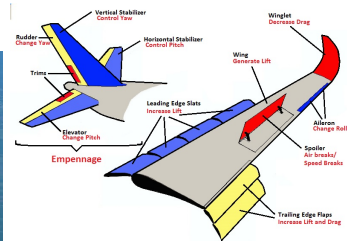
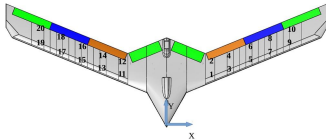
Aerospace Examples

■ B2 Stealth Bomber:

- flying wing to reduce radar signature
- hard to control: many actuators (control surfaces), each has small effect

■ Harrier jump jet:

- adjustable actuator through thrust vectoring, allows vertical and short take-off and landing (VSTOL)
- hard to control: highly nonlinear plant, multiple inputs and outputs (MIMO)



Control System Design in the AI/ML Era

- Very powerful tools!!
- Amazing success: especially in Robotics
- How do they work?
 - 1 Build a model which will serve as the "truth model"
 - first-principles: use complex dynamical models, or ...
 - data-driven methods: use enormous corpi of empirical data from experiments ...
 - 2 Carefully define performance metric
 - 3 Explore the control design space
 - use lots of simulations of the truth model under many disturbance conditions
 - disturbances include actions of exogenous human agents
 - assess performance
 - learn best control strategy

Control System Design in the AI/ML Era ...

- Some methods for systematic exploration:
 - reinforcement learning
 - imitation learning
 - "mixture of experts" learning
- Trial and error approach is fine but
 - What about safety-critical systems?
 - What about unmodeled or unsimulated environmental conditions?
 - Need some robustness and safety guarantees ...
- AI/ML are not magic wands
- Still need to understand and have engineering skills
 - to build models
 - to simulate models efficiently
 - to design control architecture